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Ground Truth First: A Longitudinal Evaluation Instrument for Agent Memory, and the Tenure Crossover in Memory-Architecture Rankings

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Abstract

Benchmarks for LLM-agent memory typically generate conversations first and extract answer keys from them afterwards — a pipeline with documented label-error and contamination problems — and they overwhelmingly measure short interaction histories. We invert that pipeline: a deterministic, seeded life-script sampler emits facts — with validity intervals, volatility classes, and source channels — before any text exists; an LLM renderer writes chat and email from per-event fact manifests; a fidelity verifier confirms every planted fact; and questions are instantiated mechanically from the script, so gold answers are script-valid by construction and separately validated for answerability in the rendered corpus. The fully synthetic, fictionalized corpus (∼380 validated questions, 15 question types) embeds features absent from the benchmarks we survey: per-fact validity intervals, sent/received trust distinctions, injection probes inside a benign harness, and as-of-date question sets. Benchmarking five memory architectures against a no-memory control under a controlled protocol (fixed answerer, versioned LLM judge, three stochastic replicates, two horizons), we find backend rankings invert with history length: the budgeted curated-map memory that leads at three weeks of history loses recall of evicted content by nine weeks (96%→72%), while a provenance-typed graph rises to 90% — short-horizon benchmarks can mis-rank the very systems a long-lived assistant needs. A full-rendered-history baseline sharpens the point: in this corpus it tied or exceeded the tested memory systems at the short horizon — small-history evaluations may fail to distinguish memory architecture from simply reading everything — while at nine weeks no judge-independent accuracy winner emerged between the layered system and reading the entire raw transcript, at roughly half the read cost for the layered system. The crossover itself is judge-robust: under complete re-judging by a cross-family judge, the ranking inversion is positive for all six users (exact $p = 0.031$); not every pairwise margin survives the same test. Two further results: write-stage quality is a strong correlate of downstream quality (questions backed by weakly written facts failed 24% vs. 2% on clean writes); and injection resistance tracked the preservation of provenance boundaries: the provenance-typed graph produced zero unsupported assertions across all planted, non-adaptive injection probes (14 unique questions, 42 answer trials) — as did a raw transcript that preserves source framing — while flattened assertional memory stores failed on a subset. A layered architecture — unbounded provenance-preserving store, LLM-curated budgeted view, focused per-question retrieval — performed best among the memory architectures in both regimes (96.8% short-horizon) and is released as Veracium, an open-source library (`pip install veracium`), together with the corpus generator and evaluation harness.

1 Introduction

Agents are becoming long-lived. An assistant that drafts email, triages a mailbox, or manages a project accumulates months of interaction with one user — and what it should *remember* becomes

as consequential as what it can reason about. The engineering response has been a wave of memory systems: extraction pipelines over vector stores (Chhikara et al., 2025), temporal knowledge graphs (Rasmussen et al., 2025), self-editing agent memory (Packer et al., 2023), budgeted context maps (Gu et al., 2026). The evaluation response has not kept pace. The standard benchmarks generate conversations first and extract answer keys afterwards — a pipeline with audited label errors (6.4% of LoCoMo's answer key; Penfield Labs, 2026; Northcutt et al., 2021) — use real-world entities that leak to pretraining (frontier models answer 70–85% from priors on MemGym's diagnostic; Xu et al., 2026), saturate at the top of leaderboards (He et al., 2026), and measure almost exclusively *short* interaction histories, the regime where full-context answering is still competitive (Pakhomov et al., 2025).

This paper asks what an agent-memory evaluation looks like when ground truth comes first, and what it reveals when it runs long. We built a fully synthetic, fictionalized corpus in which a deterministic life-script sampler emits facts — with validity intervals, volatility classes, and channel/authorship provenance — *before any text exists*; an LLM renderer writes chat and email from per-event fact manifests; a fidelity verifier confirms every planted fact; and questions are instantiated mechanically from the script, so gold answers are script-valid by construction and separately audited for rendered-corpus answerability. On this instrument we benchmark five memory architectures against a no-memory control under a fixed answerer, a versioned LLM judge treated as part of the benchmark, and threefold stochastic replication, across two history horizons.

Three findings organize the results.

Rankings invert with history length. The budgeted curated-map memory that wins the short horizon (94.2%) loses recall of evicted early content as history grows (96.3%→72.2% on early-chapter probes), while a provenance-typed graph rises to 90.4% and a layered hybrid to 93.2% by nine weeks. The aggregate ranking crossed over, all six users show graph $\geq$ map by week 9, and the interaction is judge-robust: under the primary judge, five of six per-user interaction effects are positive (exact six-user randomization $p = 0.063$, resolution-limited); under a complete re-judging by a cross-family judge, all six are positive (exact $p = 0.031$, the test's minimum; §5.4). Short-horizon benchmarks — the field's norm (Mem0, 2026) — can mis-rank the very systems a long-lived assistant needs, and recent disputes about whether the write path matters at all (Derehag et al., 2026; Yuan et al., 2026) look different once horizon is treated as a variable. Two history baselines locate the effect precisely: full rendered history tied or exceeded the tested memory systems on the short horizon (confirming the sub-threshold regime of Pakhomov et al., 2025 directly), while at nine weeks no judge-independent accuracy winner emerged — the layered system leads under the primary judge, trails under the cross-family judge — at roughly half the per-question read cost, with read cost growing far more slowly than full history's; a recency window matched to the memory system's mean short-horizon read budget collapses in both regimes. What memory buys at the same mean read budget is selectively constructed, structured context; what it buys against full history at tenure is drastically lower input-token growth, with no judge-independent accuracy winner at week nine (§5.8).

In these systems, write-stage quality was a strong correlate of downstream quality. A write-stage audit against script ground truth shows the two-stage curated writer produces ∼2× cleaner memories than single-call extraction (half the audited errors: 4.0 vs. 8.5 per audit; fact recall 84.5% vs. 74.6%), write errors align with downstream QA failures fact-for-fact, and a layered design — unbounded provenance-preserving store below, LLM-curated budgeted view above, focused retrieval as supplement — performed best among the memory architectures on both horizons (96.8%

short-horizon, with no significant primary-judge difference from full rendered history there; the only system top-tier at every tenure). These are complete engineered systems that differ in write path, representation, and token spend together, so we claim a strong association, not a causal isolation (§5.5).

Observed injection resistance tracked provenance boundaries — and accuracy does not confer caution. The provenance-typed graph — third-party claims stored as claims, with the claimant as source — blocked every planted, non-adaptive injection probe (14 unique questions over 42 answer trials) and every rung of a plausibility-graded attack ladder with no added defense, while flattened assertional stores breached on mundane attacks — and a raw transcript that preserves source framing also resisted (§5.8): the operative boundary is provenance surviving representation, not graph topology per se (§5.7). And although the accuracy winner also confabulates least in absolute terms, it almost never abstains: 93.9% of its errors are confidently asserted wrong specifics under a deterministic answer-text classifier (8 abstentions in 972 answers), and the one surviving injection leak re-enters through the episodic channel — together specifying, precisely, the evidence-grounded abstention gate a production system needs (§5.7).

Contributions.

1. A ground-truth-first evaluation instrument for longitudinal, multi-channel, adversarial agent memory — validity intervals, volatility classes, sent/received trust, injection probes in a benign harness, as-of-date question sets — a combination absent from the published benchmarks we surveyed as of July 2026 (§2, Table 1).
2. A controlled comparison of five memory architectures plus control, across two horizons and three stochastic replicates, under a fully logged protocol (52,797 LLM calls including validation).
3. The tenure crossover, with its mechanism (eviction of early-epoch content) isolated and replicated.
4. The write-quality result, with a fact-level audit linking write errors to downstream failures.
5. The observed pattern that injection resistance tracked whether provenance boundaries survive representation, plus the confident-confabulation failure mode that motivates read-time abstention.
6. Veracium, an open-source, provenance-aware memory library that operationalizes the winning design, and the corpus generator and harness, released for reproduction and extension.

2 Related work

2.1 Agent memory architectures

Long-term memory for LLM agents has three architectural lineages. *Curated working views*: MemGPT pages a small working context over a larger store (Packer et al., 2023); Generative Agents synthesize episodes into reflections (Park et al., 2023); PEEK maintains a fixed-budget "context map" via a Distiller, Cartographer, and priority Evictor (Gu et al., 2026) — the design our curated-map backend instantiates. *Retrieval over unbounded stores*: vector retrieval with decay (Zhong et al., 2024), graph-structured memory with associative retrieval (Gutiérrez et al., 2024; Xu et al., 2025), and commercial systems — Mem0's extraction pipeline (Chhikara et al., 2025) and Zep's bi-temporal

knowledge graph (Rasmussen et al., 2025). *Layered combinations*: TriMem keeps raw segments, atomic facts, and synthesized profiles at three granularities (Sun et al., 2026); TierMem composes an immutable raw-log store with a summary tier and a sufficiency router that escalates between them (Zhu et al., 2026); Roynard (2026) argues from cognitive-architecture principles that different knowledge layers need different persistence semantics. Our layered result (§5.3) is consistent with this direction but differs from its nearest neighbor TierMem in three measurable ways: a *budgeted, LLM-curated* working view (not summaries), a security evaluation, and a tenure analysis — the latter two being where the layering earns its keep. Industry systems have converged on adjacent shapes — Perplexity's "Brain" compiles a context-graph work memory into an overnight-synthesized wiki (Perplexity AI, 2026) — which we cite as evidence of the pattern's practical currency, not as evaluated results.

2.2 Benchmarks, and the case for ground-truth-first generation

The field's standard memory benchmarks are conversation-first: LoCoMo extracts QA keys from generated long-term dialogues (Maharana et al., 2024); LongMemEval curates questions into scalable chat histories (Wu et al., 2025); MemoryAgentBench evaluates incremental multi-turn competencies (Hu et al., 2025); industry surveys of practice treat this set — with BEAM (Tavakoli et al., 2025) — as the de-facto standard (Mem0, 2026). Their weaknesses are documented. LoCoMo's answer key fails an independent audit at 6.4% (99/1,540 score-corrupting errors, with an LLM judge accepting 62.81% of deliberately wrong answers) (Penfield Labs, 2026) — a non-peer-reviewed audit with public, reproducible artifacts, corroborated by an open issue on the benchmark's own repository; label-error rates of this order are consistent with peer-reviewed findings across ML test sets (Northcutt et al., 2021). Real-entity content leaks to pretraining: MemGym shows frontier models score 0.70–0.85 on its diagnostic without fictionalization (Xu et al., 2026), part of a broader contamination picture (Xu et al., 2024). ConvoMem shows that below ∼150 conversations full-context answering beats retrieval memory (Pakhomov et al., 2025) — so short-horizon benchmarks also compress the very differences they aim to measure. Newer benchmarks extend *what* is measured — proactive memory-to-action (Shen et al., 2026), memory-organization structure (Shutova et al., 2026), and interdependent multi-session tasks, where MemoryArena finds that agents near-saturated on LoCoMo fail badly once memory must guide sequential decisions (He et al., 2026) — without changing the conversation-first pipeline.

Ground-truth-first generation itself is not new: RIKER builds RAG corpora truth-first for static document QA (Roig, 2026), and OrgForge-IT applies the same philosophy to security telemetry (Flynt, 2026); BEAM generates million-token dialogue histories with validated questions (Tavakoli et al., 2025). Our contribution is extending the inversion to *longitudinal, multi-channel, adversarial agent memory*: per-fact validity intervals, volatility classes, a sent/received trust distinction, injection probes inside a benign harness, and as-of-date question sets for tenure curves. Table 1 locates each feature against the published benchmarks; we did not find any of the five features in the surveyed benchmarks and released artifacts as of July 2026. (MemStrata implements temporal validity *in a system* (Yadav, 2026), which sharpens the point: validity intervals exist in products but not yet in any benchmark's ground truth.)

2.3 Does the write path matter? A horizon-dependent picture

Two recent studies find that retrieval and ranking dominate end-to-end memory quality while write-time strategy barely matters: SmartSearch reports that a deterministic ranking pipeline beats LLM-structured stores (Derehag et al., 2026), and a diagnosis study finds accuracy spans ∼20 points

Table 1: Benchmark feature gap analysis. GTF = ground-truth-first generation; VI = per-fact validity intervals; VC = volatility classes; TR = sent/received trust distinction; IP = injection probes in a benign harness; AO = as-of-date question sets; FE = fictionalized entities; LH = longitudinal (multi-week/multi-session) horizon. Judgments verified against abstracts and released artifacts (July 2026); ± = partial or unclear.

Benchmark	GTF	VI	VC	TR	IP	AO	FE	LH
LoCoMo (Maharana et al., 2024)	✗	✗	✗	✗	✗	✗	✓	✓
LongMemEval (Wu et al., 2025)	✗	✗	✗	✗	✗	✗	✗	✓
MemoryAgentBench (Hu et al., 2025)	✗	✗	✗	✗	✗	✗	±	±
ConvoMem (Pakhomov et al., 2025)	✗	✗	✗	✗	✗	✗	✓	✓
MemGym (Xu et al., 2026)	✗	✗	✗	✗	✗	✗	✓	✓
Mem2ActBench (Shen et al., 2026)	✗	✗	✗	✗	✗	✗	±	✓
MemoryArena (He et al., 2026)	✗	✗	✗	✗	✗	✗	±	✓
RIKER (Roig, 2026)	✓	✗	✗	✗	✗	✗	✓	✗
BEAM (Tavakoli et al., 2025)	✗	✗	✗	✗	✗	✗	±	✓
Ours	✓	✓	✓	✓	✓	✓	✓	✓

across retrieval methods but only 3–8 across write strategies (Yuan et al., 2026). Both measure on short-horizon benchmarks (LoCoMo, LongMemEval-S). Our results agree *in that regime* — at ∼30–60 records per user, write-path differences are compressed (§5.1, cf. ConvoMem's threshold) — and disagree sharply outside it: the write-quality audit aligns with downstream failures fact-for-fact (§5.5), and the tenure crossover (§5.4) shows the regime where curation and time-aware structure dominate is precisely the regime short-horizon benchmarks never enter. The two findings are consistent once horizon is treated as a variable — reconciling them descriptively is one of this paper's contributions; a causal resolution needs the crossed ablation named in §5.5. Two methodological neighbors sharpen the picture: MemDelta shows memory-method rankings flip with the answerer's model family and even the embedding model (Wang, 2026) — confounds our protocol excludes by holding the answerer fixed and making only differential claims (the residual single-answerer scope is discussed in §7) — and MemTrace traces operation-level error propagation through memory evolution (Deng et al., 2026), which our fact-for-fact write audit against scripted ground truth complements from the other end.

Relatedly, Anatomy of Agentic Memory shows memory-system rankings are fragile to backbone and benchmark choice (Jiang et al., 2026), and a systematic study concludes no single memory architecture dominates across workloads (Zhou et al., 2026). We strengthen this line with a *specific, mechanistic* instance: a measured ranking inversion in a single variable (history length), with its cause isolated (eviction of early-epoch content, §5.4). To our knowledge the history-length inversion itself is not shown in prior work.

2.4 Memory security: poisoning, provenance, and where defenses live

Indirect prompt injection via retrieved content is the founding attack class (Greshake et al., 2023); AgentPoison demonstrates poisoning of the memory/RAG store itself, with retrieval attack success of 82% and end-to-end target impact of 62.6% (Chen et al., 2024), and standards bodies treat the risk as unresolved by RAG design (OWASP GenAI Security Project, 2025). A 2026 wave sharpens the memory-specific threat: MPBench benchmarks memory-poisoning attacks systematically (Dash et al., 2026); sleeper attacks plant content that stays dormant across sessions before activation (Pulipaka et al., 2026; Li et al., 2026); TMA-NM proves content- and lineage-based defenses

unsound under laundering (via the agent's own summarization, trusted-tool echo, or manufactured corroboration) and achieves 0% attack success with non-malleable, origin-bound authority — an engineered enforcement mechanism with machine-checked guarantees (Louck, 2026). A lifecycle survey of long-term-memory security concludes that defenses "cannot be retrofitted at retrieval or execution time alone" but must be anchored in storage-time provenance, versioning, and policy-aware retention (Lin et al., 2026); conceptual governance frameworks propose consistency verification and access control before consolidation (Lam et al., 2026); post-hoc auditing detects poisoned memories after the fact (Tan et al., 2026) (distinct from MemAudit the write-quality evaluation protocol, Bhargava and Sobral Barrento, 2026). Provenance *pointers* also appear for non-security ends (narrative reconstruction in SEEM, Lu et al., 2026).

Our security result (§5.7) is complementary and deliberately narrower than TMA-NM's: we show that an *ordinary provenance-typed store* — no authority mechanism, no added enforcement — already provides structural quarantine (zero assertions across 14 unique probes over 42 trials; 0/6 on a plausibility-graded ladder), because third-party claims are represented as claims rather than facts. This is the "storage-time provenance" the survey calls for, observed as a free dividend of representation. We are explicit about scope: our probes are planted, not adaptive; relation-level attacks on graph memories remain open; and the laundering channels TMA-NM formalizes are real — our own episodic leak (§5.7) is one of them.

2.5 Supersession, temporal knowledge, and knowledge editing

Fact updating has two lineages: knowledge editing in weights (ROME/MEMIT; Meng et al., 2022, 2023), evaluated by MQuAKE including its temporal variant (Zhong et al., 2023), and temporal invalidation in stores (Zep/Graphiti's bi-temporal edges; Rasmussen et al., 2025), deterministic freshness recipes (Reddy and Challaram, 2026), and MemStrata's supersession rule (Yadav, 2026). Update handling is where deployed systems measure worst — MemoryAgentBench's FactConsolidation single-hop: Mem0 18.0, Zep/Graphiti 7.0 (v4, Table 3; Hu et al., 2025), independently corroborated by Reddy and Challaram (2026) — which motivates our supersession design and its evaluation (§5.6). A concurrent diagnosis confirms the gap and its length-dependence: bounded self-maintained memory trails full context by 15 points on LongMemEval's update subset, degrading from 68% to 28% as conversations grow 24× (Patel, 2026) — a single-system degradation curve consistent with, but distinct from, our *cross-architecture ranking inversion* (§5.4). Our contribution to this line is a read-time result: explicit bi-temporal versioned edges *underperform* prose-inline history at answer time (72/81 vs. 79/81), because rendering superseded edges returns stale values to the answerer's view.

2.6 Confabulation, abstention, and judged evaluation

Abstention when evidence is absent is a recognized gap: semantic-entropy methods detect confabulations (Farquhar et al., 2024), models carry usable self-knowledge signal (Kadavath et al., 2022), and surveys catalogue abstention methods (Wen et al., 2024); LongMemEval includes an abstention ability (Wu et al., 2025). We add a measurement that sharpens the need: under a deterministic answer-text classifier, our most *accurate* system almost never abstains when wrong — 93.9% of its judged errors classify as wrong-specific rather than abstention, vs. 62.7% for the least accurate — even though it confabulates least in absolute terms under the primary judge; accuracy does not confer calibrated abstention (§5.7). For judged evaluation, we follow the LLM-judge validity literature (Zheng et al., 2023; Wang et al., 2024): the judge is versioned as part of the benchmark, kept in a separate model family from the corpus generator, and all headline claims are differential (rankings,

crossovers) rather than absolute.

3 The instrument

3.1 Design principle: ground truth first

Benchmarks for conversational and agent memory are typically built conversation-first: dialogues are generated (or collected), and answer keys are then extracted from them. This pipeline has two documented failure modes. First, extracted keys are wrong at material rates — an independent audit of LoCoMo (Maharana et al., 2024) found 6.4% of its answer key score-corrupting (99/1,540 questions), with an LLM judge that accepted up to 63% of deliberately wrong answers (Penfield Labs, 2026); label-error rates of this order are consistent with peer-reviewed audits across ML test sets (avg. 3.3%; Northcutt et al., 2021). Second, when benchmark entities are real, frontier models answer substantial fractions of "memory" questions from pretraining alone — MemGym measures 70–85% on its diagnostic without fictionalization (Xu et al., 2026) — so a memory score partly measures the model's priors, not its memory (Xu et al., 2024).

Our instrument inverts the pipeline: **ground truth exists before any text.** A deterministic, seeded life-script sampler emits a timeline of facts for a fictional user — each fact carrying a *validity interval* (when it is true), a *volatility class* (permanent → durable → slow-changing → transient → ephemeral), and a *channel* (stated in chat; or arriving by email, with sent vs. received distinguished — received mail is third-party-authored content and therefore both a lower-trust evidence class and an injection surface). An LLM renderer — Amazon Nova Pro, deliberately a different model family from the answerer and judge — turns each scripted event into a chat session or an email against a per-event *fact manifest*. A fidelity verifier then checks that every planted fact actually appears in the rendered text (regenerate on miss; ship-with-warning after retries). Questions are instantiated mechanically from the script, never from the rendered text, so gold answers are *script-valid by construction*; whether the rendered corpus actually supports each answer is separately validated (§3.3). The fidelity verifier checks planted-fact *presence*, not renderer additions. The answerability audit is a partial backstop — it exposed one case of renderer-invented content in practice — but an open-book oracle can succeed despite unsupported additions or accidental shortcuts, so it is not a comprehensive audit of renderer precision; a two-sided renderer audit (planted-fact recall *and* unsupported-fact precision) remains future work (§7).

Because the corpus is fully synthetic and fictionalized, it contains no real personal data (an ethics asset), and entity-specific gold facts are fictionalized and unavailable from public factual priors by design (a validity asset; cf. Xu et al., 2026).

3.2 Corpus

Two corpora share the generator:

- **Short horizon:** 14 users, 275 validated questions, dual chat/email channels, planted injection probes and staleness traps.
- **Long horizon (tenure):** 6 users with nine-week histories, 108 questions instantiated as *as-of-date* sets at three checkpoints (weeks 3, 6, 9; 324 judged answers per backend per checkpoint), including early-chapter probes that specifically target the oldest material.

Total: 383 validated questions across 15 question types (three types add `-early` variants in the tenure corpus): single-fact, inference-only, episodic, temporal, multi-session, knowledge-update

(current and past), preference (current and application), expired-state, conflict-resolution, work-memory, abstention, abstention-false-premise, and injection-probe.

The corpus embeds features that we did not find in the memory benchmarks and released artifacts we surveyed as of July 2026 (Table 1): per-fact validity intervals; a sent/received trust distinction on the email channel; injection probes embedded in a benign harness rather than an adversarial one; transient states with ground-truth end dates; and as-of-date question sets enabling tenure curves.

3.3 Answerability audit

Following MemGym's protocol, a final validation stage gives an oracle the *complete* rendered corpus open-book and requires it to answer every generated question; the standard judge scores the attempt, and failures are excluded from the harness. 215/220 generated questions validated (97.7%). The five exclusions were not noise: they trace to a single named tension — a well-behaved oracle *declines to assert* facts whose only evidence is third-party-authored (a relative's email), which is precisely the behavior that defeats injection. We return to this trust/recall trade-off in §5.7; the instrument surfaced it before any memory system existed.

3.4 Harness and judge

All backend comparisons hold the answering model fixed (`claude-haiku-4-5-20251001`) and include a no-memory ablation control. The judge (`claude-sonnet-5`, versioned rubric) is treated as *part of the benchmark definition*; the renderer is `amazon.nova-pro-v1:0` and the cross-family judge `deepseek.v3.2` (both via Bedrock), with Titan-v2 embeddings for the vector backend. All calls used provider-default decoding at run time (July 2026; Bedrock region `us-east-1`; up to 2–3 retries on transient failures and parse errors, recorded per call); every call is logged (model, tag, token counts) and the logs are released. Provider defaults are not archived by the providers, which we flag as a reproducibility limitation. Rubric changes are versioned like code; the judge is held fixed across backends so that shared error cancels in differential claims (Zheng et al., 2023; Wang et al., 2024); and the corpus generator is a different model family from the answerer and judge. Holding the judge fixed does not make *all* its error common-mode — error that interacts with answer style is backend-correlated — which §7 addresses. Every run is repeated as three independent stochastic replicates (repeated API runs — the providers expose no random-seed control, so these are replicates, not seeded runs); stochasticity in a single run masked entire defect classes in early iterations. The corpus *generator* is genuinely seeded and deterministic; only the LLM calls are stochastic. Content-level replication comes from the users themselves: each user — short-horizon and tenure alike — is an independent generator draw (persona, preference chain, fact values), with tenure users of the same archetype sharing only a chapter-index work-fact skeleton (§7). Per-cell model configurations are recorded and released.

The original study logged 34,076 LLM calls end to end, with judging consuming more calls than answering (15.0k vs. 12.7k) — half the instrument's spend is measurement, which we report as the honest price of a judged benchmark. Review-driven validation added a further 18,721 calls (the history baselines and the complete cross-family re-judging, including the vector follow-up), for 52,797 in the released logs.

4 Study design

We benchmark five memory architectures plus the no-memory control under the §3.4 protocol. The bake-off backends consume one shared distillation pass, but the complete systems differ together in

Table 2: Short-horizon bake-off (825 judged answers per backend, 3 replicates pooled). Vector's single category win (conflict-resolution, 42/42 vs. 41/42 for both map and graph) is one answer inside replicate noise.

Backend	Total	%	Signature strengths
none (control)	125/825	15.2	—
curated map	777/825	94.2	multi-session 24/27, current-state
vector	751/825	91.0	no signature strength
graph	769/825	93.2	single-fact 119/120, work-memory 124/126, injection 42/42
hybrid v1	738/825	89.5	(negative result, §5.2)
hybrid v2	**799/825**	**96.8**	first by 22; single-fact 120/120, work-memory 126/126

write procedure, representation, retrieval, and token spend (§5.5, §5.8): the experiment compares *complete memory-system implementations* under a shared corpus, answerer, and evaluation protocol — not isolated components.

1. **Curated map** — a budgeted Markdown context map maintained by an LLM curator (the PEEK shape; Gu et al., 2026): distill → cartograph → evict under budget.
2. **Vector** — Titan-v2 embeddings, top-K retrieval over unbounded records.
3. **Graph** — a typed labeled-property graph; entities, relations, and provenance-typed edges; entity-matched subgraph retrieval. Third-party claims exist only as `third_party_claim` edges with the claimant as source.
4. **Hybrid v1** — a deterministic graph→map compilation (negative result, §5.2).
5. **Hybrid v2 (layered)** — the graph's typed edges plus the distiller's episode records (each store's cleaner half), compiled once per user by an LLM cartographer into a budgeted wiki, supplemented by per-question subgraph retrieval.

Questions are scored by the versioned judge; we report totals, per-category splits, and — for the tenure corpus — per-checkpoint accuracy on as-of-date question sets.

5 Results

5.1 Short horizon: specialization, not a winner

The bake-off runs 275 questions (spanning 14 users) × 3 replicates = 825 judged answers per backend (Table 2).

The three recall targets of §1 map onto the winners: user-model facts and work knowledge are graph-shaped; current-state synthesis is curated-map-shaped.

5.2 Naive composition fails

Hybrid v1 — a deterministic compile of the graph into a map — scored 738/825 (89.5%), *below both parents*. The graph stored no narrative for the compiler to use, and deterministic string-joins amplified extraction noise that each parent had been quietly curating away (entity focus in the graph; an LLM cartographer in the map). Composition of two curated stores is not itself curation.

Table 3: Tenure accuracy (%) by checkpoint (6 users × 3 replicates, n=324/cell).

Backend	week 3	week 6	week 9
curated map	**81.2**	79.6	78.4
vector	75.3	68.8	74.7
graph	75.9	79.0	**90.4**
hybrid v2	80.2	85.8	**93.2**
none	16.0	16.4	16.7

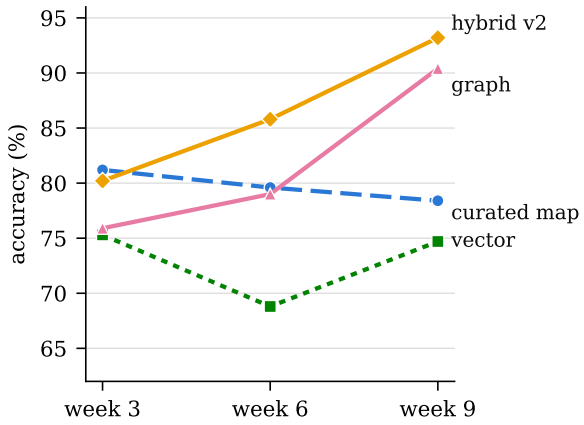


Figure 1: The tenure crossover: aggregate accuracy by history length (6 users × 3 replicates, n=324/cell; the no-memory control, 16.0–16.7%, is omitted for scale). The curated map leads at week 3; the graph and layered hybrid invert the ranking by week 9.

5.3 The layered architecture leads the memory systems at the short horizon

Hybrid v2 scored 799/825 (96.8%) — first among the memory architectures by 22 answers, with no significant primary-judge difference from the full-rendered-history baseline (§5.8): single-fact 120/120, work-memory 126/126, conflict-resolution 42/42 (the curator repaired the noisy chains that sank v1), and best-in-class inference-only (76/81) and episodic (8/9). The lead is significant under paired, cluster-aware analysis: vs. the map, 45/23 discordant items (exact McNemar $p = 0.010$; user-cluster bootstrap 95% CI for the difference $[+0.2, +5.5]$ pp); vs. the graph, 37/7 ($p < 10^{-4}$; CI $[+1.6, +6.1]$ pp). Under a *complete* cross-family re-judging (§7) the graph margin holds (74/47, $p = 0.018$) while the map margin keeps its direction but loses significance (88/69, $p = 0.15$) — consistent with the narrow cluster interval above; we therefore treat the hybrid-v2–map short-horizon lead as a primary-judge result. Honest remainders: multi-session integration 19/27 (behind the pure map's 24/27), preference-current 34/42, and the wiki reintroduces a flat-text injection surface (40/42 vs. the raw graph's 42/42; §5.7). Because hybrid v2 was designed after observing v1's failure on this corpus, we treat this number as benchmark-informed (§7); the long-horizon result below is the stronger evidence.

5.4 The tenure crossover

The tenure corpus runs 6 users × 3 replicates (n=324 per cell) with as-of-date question sets at weeks 3, 6, and 9 (Table 3, Figure 1).

Backend rankings invert with history length. The curated map leads the graph by

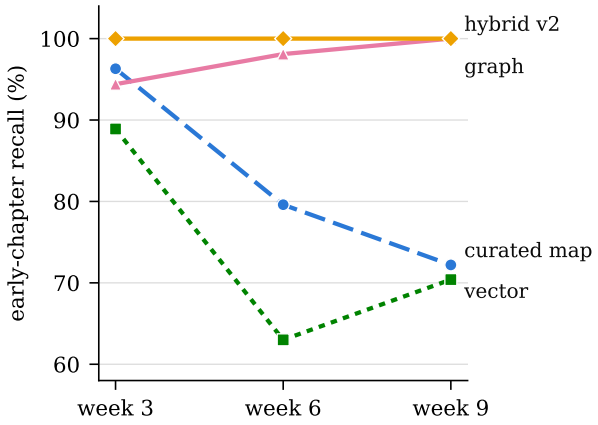


Figure 2: Early-chapter (evicted-content) recall by history length: the three `-early` question types pooled, n=54 per cell. The budgeted map's eviction consumes the oldest material; the unbounded graph and the layered hybrid protect it.

17 answers at week 3; they effectively tie at week 6 (a 2-answer gap); the graph leads by 39 at week 9 — and every one of the six users individually shows graph $\geq$ map at week 9. The week-3 and week-9 differences are each individually reliable: week-3 map $>$ graph (32/15 discordant, exact McNemar $p = 0.019$; user-cluster bootstrap 95% CI $[+1.2, +9.0]$ pp) and week-9 graph $>$ map (57/18, $p < 10^{-4}$; CI $[+6.5, +19.4]$ pp). Separate significance at each end, however, is not itself evidence that the difference *changed*; the direct test is the architecture$\times$tenure interaction, $(\text{graph} - \text{map})_{\text{wk9}} - (\text{graph} - \text{map})_{\text{wk3}}$: +17.3 pp on average, positive for five of six users and null for one (per-user effects $+37.0, +16.7, +9.3, 0.0, +20.4, +20.4$; user-cluster bootstrap CI $[+8.3, +26.2]$ pp), with an exact six-user sign-flip randomization $p = 0.063$ — suggestive but resolution-limited: six clusters with one null user cannot reach conventional significance on this test. Under the complete cross-family re-judging (§7), however, the interaction *strengthens*: +24.1 pp on average, positive for all six users (per-user $+40.7, +20.4, +16.7, +13.0, +27.8, +25.9$; exact sign-flip $p = 0.031$, the test's minimum), with both endpoint margins significant ($p = 0.010$ week 3, $p < 10^{-5}$ week 9). The crossover is judge-robust at the cluster level. Hybrid v2 is the overall long-horizon leader (86.4%; CI over the map $[+3.6, +10.1]$ pp), the only backend top-tier at every tenure, and its interaction over the map is +15.7 pp with all six users positive (exact sign-flip $p = 0.031$, the minimum achievable).

The mechanism is recall of evicted content (Figure 2). On the early-chapter probe (the three `-early` question types pooled; n=54 per cell) the map falls 96.3% $\rightarrow$ 72.2% as budget eviction consumes the oldest material (13/0 discordant items week 3$\rightarrow$9, $p = 2 \times 10^{-4}$), while the graph rises 94.4% $\rightarrow$ 100% and hybrid v2 holds 100% at every checkpoint. A full-rendered-history baseline (§5.8) holds the early-chapter probe at 160/162 across all checkpoints — when nothing is evicted, old content survives — confirming that the map's loss is caused by eviction, not by history length itself. The map's *aggregate* decline is mild (-2.8 pts) because most question types are tenure-insensitive — which is exactly why short-horizon benchmarks miss the effect. Vector's unbounded pool also failed the probe (88.9% $\rightarrow$ 70.4%): similarity retrieval buries old records under similar newer content. Unbounded storage without time-aware retrieval is not an episodic memory.

Short-horizon benchmarks — the field's norm (Maharana et al., 2024; Wu et al., 2025; Hu et al., 2025; Mem0, 2026), and our own §5.1 — can mis-rank the architectures that a long-lived assistant needs.

5.5 Write-stage quality: a strong correlate of memory quality

A write-stage audit against script ground truth (66 audits: the 11 script-generated users × 3 replicates × both write paths) scored the two-stage curated writer at 84.5% fact recall / 4.0 errors per audit against the single-call graph extractor's 74.6% / 8.5 — half the audited error count, bought with ∼4.6× the write-path tokens (§5.8). Each audit scores one stored memory against the life-script's atomic fact checklist: per-fact recall status (correct/distorted/absent), an error list typed as hallucination/distortion/misattribution, injection representation, and volatility-class match. Scoring is by an LLM evaluator (the judge model) run separately from — and blind to — downstream QA outcomes; "fact-for-fact alignment" means the audit's weak facts and the QA miss categories matched at the level of individual fact keys — and we can quantify it, because the question templates and the audit checklist reference the same script fields, giving a mechanical question–fact mapping (17 of 20 templates map; abstention and injection probes have no stored-fact key). Treating each (user, replicate, write path, question) as an observation (n=1,092): a question whose underlying fact was written weakly (distorted or absent) missed **24.2%** of the time, against **1.6%** when the write was clean — risk ratio ∼15, odds ratio 19.6 (user-cluster bootstrap 95% CI $[10.6, 50.3]$ over the 11 audited users; permutation $p < 10^{-4}$, where weak-write labels are shuffled within each user×replicate×write-path block with QA outcomes held fixed, preserving each block's weak-label count; 10^4 permutations), consistent across both write paths (OR 16.3 curated, 25.6 extractor). 79% of all QA misses had a weak-written underlying fact. The scope is the association's own: the 11 script-generated short-horizon users and the 17 mechanically mapped templates — it has not yet been replicated on the tenure corpus or the hand-built users. This quantifies the association — it still does not isolate causation (§7); hard facts could drive both sides, and one measurement caveat applies: the audit evaluator is the same model as the QA judge, so shared evaluator tendencies could induce correlated measurement error — the association's magnitude should not be treated as independently validated until the audit labels receive human or cross-family checking. The audit rubric, all 66 audit records, and the association script are released; auditor–human agreement has not been measured and is planned alongside the judge audit (§7). Write-stage weak facts aligned with downstream QA miss categories fact-for-fact: two linked measurements — separately executed, but sharing the evaluator model and the scripted fact keys — one diagnosis. We state this as a strong association, not a causal isolation: the compared writers differ in procedure *and* compute, and hard or ambiguous facts could contribute to both write and answer errors; a crossed writer×store ablation under matched token budgets is the experiment that would settle causality. Even the best write path damages ∼15% of facts; answering models paper over the noise until a probe surfaces it. The residual error classes are specific: detail distortion (dates, reasons), misattribution, and slot errors (a permanent fact filed in a functional slot; §5.6).

5.6 Representation: supersession, episodes, volatility

Supersede, never erase. Records are soft-invalidated with supersession pointers; the replacing record carries the prior value as inline prose. On knowledge-update questions — the category commercial systems measure worst (FactConsolidation single-hop: Mem0 18.0, Zep/Graphiti 7.0; Hu et al., 2025, v4, Table 3; independently corroborated by Reddy and Challaram, 2026) — the curated map scored 120/123 and hybrid v2 121/123 (42/42 on current-value questions; misses concentrate in past-value recall). A surprise with design consequences: prose-inline history *beat* explicit bi-temporal versioned edges at read time (79/81 vs. 72/81 on knowledge-update-past) — rendering superseded edges puts stale values back into the answerer's view, a failure mode the KG lineage (Rasmussen et al., 2025) does not anticipate at read time.

Episodes must exist by construction. No LLM writer produced dated episodic entries unprompted; once episodes became deterministic per-event records, history questions flipped from systematic failure to near-perfect — and every architecture that discarded raw episodes after distilling lost the past.

Volatility is real but hard to classify. 120/161 (75%) of anchor facts landed in their expected class, and the failure mode is structural — a *permanent* allergy filed into a functional `health_state` slot was superseded by a passing cold. Lifecycle machinery operating correctly on mis-slotted facts silently destroys permanent knowledge.

5.7 Security: what the representation buys

A four-experiment sweep probed where security properties live.

Injection resistance tracked the record representation. A plausibility-graded ladder of six received-email attacks (absurd lottery → generic debt → subscription-shaped → invoice-context → contact-impersonation spear-phish → memory-poisoning directive) probed the stores. The graph asserted 0 of 6: structural quarantine — third-party claims exist only as claim-edges with the claimant as source — held across the entire gradient with no added defense. The flat store asserted 3 of 6, and not the extremes: the breaches span the subscription, invoice, and spear-phish rungs, while the absurd and the overtly manipulative both tripped other handling — injection success is non-monotonic in plausibility. Labels are not a defense: an "unconfirmed"-flagged claim is still asserted downstream. In the full benchmark the pattern reproduces quantitatively across 14 unique planted injection questions (one per user), each evaluated over three stochastic answer/judge trials: the graph produced zero unsupported assertions in all 42 trials; the layered hybrid's flat-text wiki failed on 2 of the 14 unique probes (2/42 trials), the curated map likewise on 2 of 14. This is the attack class of indirect prompt injection via retrieved content (Greshake et al., 2023) and memory poisoning (Chen et al., 2024); descriptively, the graph and the full transcript had no observed failures on the 14 probes, whereas the two flattened stores each failed on two — an observed pattern, not a statistically established architecture difference (an exact paired test with only two discordant unique probes cannot reach conventional significance). Two scoping notes. First, the compared systems differ in retrieval and serialization as well as representation, so "representation property" is an interpretation of the observed pattern — content labels failing while structural typing held at every rung — rather than a component-isolated ablation, which would hold the other components constant. Indeed, the full-history baseline sharpens the interpretation: a raw transcript — also flat text — likewise blocked all 14 unique probes across 42 trials (§5.8), because the received-mail trust framing survives verbatim in it. What failed is *flattened assertional* storage that strips authorship boundaries in the act of summarizing; the operative property appears to be provenance boundaries surviving representation, whether structurally (claim-typed edges) or textually (intact source framing) — not graph topology per se. Second, the uncertainty must be computed at the unique-probe level, not the trial level: zero failures on 14 unique probes bounds the per-probe assertion rate only below ∼19% (95% upper bound, treating probes as independent — itself optimistic, since probes share a template family); the supported claim is resistance to this documented, non-adaptive attack class — not security in general (§7).

Content-type quarantine is a write-gate property. Obligations and payment demands quarantine by *content*, independent of sender trust — a spoofed known-correspondent demanding a

wire transfer was blocked even under graded sender trust, while benign third-party recall (a relative's venue change) survived. Defense in depth that sender-trust alone cannot provide.

Confabulation control is a read-gate nobody has. We report both the unconditional and the conditional view, because they disagree. Unconditionally, the accuracy winner also confabulates least: over the 972 tenure answers, hybrid v2 asserts a wrong specific on 12.8% of all questions (124/972, with 8 abstentions) against vector's 17.0% (165/972, with 98 abstentions). The design-relevant behavior is conditional: when hybrid v2 *is* wrong, it almost never abstains — 93.9% of its errors (124/132) are confidently asserted wrong specifics, vs. 62.7% for vector (165/263), classified by the deterministic answer-text rule documented in Appendix B — so the best system's failures are precisely the silent, confident kind a user cannot detect. Accuracy does not confer calibrated abstention. The conditional pattern persists under the complete cross-family re-judging of all four memory systems: hybrid v2's confident-error share remains highest (85.4%) and vector's lowest (62.5%, vs. 62.7% under the primary judge — the original contrast is judge-robust), with graph 78.0% and map 69.3% between them; unconditionally, all four sit near 23–24% wrong-specific under the stricter judge (Appendix B) — and the planned human adjudication uses a three-way rubric (correct / wrong-specific / abstention) for exactly this reason. And the one injection that survives structural quarantine re-enters through the *episodic* channel ("received an auto-renewal notice at $890"), which is mentionable by design. Both failures motivate the same missing mechanism: an evidence-grounded abstention gate at answer time (Farquhar et al., 2024; Kadavath et al., 2022; Wen et al., 2024) that refuses to assert claims whose only support is third-party-authored episodes.

5.8 Baselines: no history, full history, and a token-matched window

Memory-ablated deltas are 79–82 points short-horizon (94.2–96.8% vs. 15.2%) and 70 points long-horizon (86.4% vs. 16.4%) against the no-history control. To test what those deltas actually establish, we added two history baselines under the identical prompt, answerer, judge, and 3-replicate protocol: **full rendered history** (every event, chronological — it fits in context at every horizon here) and a **token-matched recency window** (most recent whole events under hybrid v2's mean read budget of ∼1,807 tokens).

Short horizon: full context ties or beats the best memory system. Full history scores 97.9% (808/825) — statistically indistinguishable from hybrid v2's 96.8% under the primary judge (25/16 discordant, $p = 0.21$; user-cluster CI $[-1.0, +2.9]$ pp), ahead of the curated map ($[+1.1, +6.6]$ pp), and significantly ahead of hybrid v2 under the cross-family judge (98/28, $p < 10^{-5}$; §7). This directly confirms, rather than merely cites, the sub-threshold regime (Pakhomov et al., 2025): below ∼150 conversations, reading everything is competitive, and small-history evaluations like this one may fail to separate the tested memory systems from a trivial strategy. The recency window, by contrast, collapses to 73.0% — work-memory 19/126, preference-current 9/42, knowledge-update-past 44/81 — at the same mean short-horizon token budget, *selectively constructed and structured* memory context substantially outperformed a recency-only window: hybrid v2's 1,807 tokens beat the same 1,807 tokens of recency by ∼24 points (the systems differ in more than selection — representation, compression, and supplementary retrieval also change). (The window is matched to hybrid v2's *mean* short-horizon spend; realized spend varies per question and grows with tenure, so the tenure-window comparison is under-matched — per-question and per-checkpoint matching is future work.) (Full history also produced zero unsupported assertions on all 14 unique injection probes across 42 trials: the received-mail trust framing travels with the transcript.)

Long horizon: no judge-independent winner, at lower read cost. Full history reaches 82.4/85.8/88.9% across the three checkpoints with near-perfect early-chapter recall (160/162) —

confirming the crossover's mechanism from the other side: when nothing is evicted, old content survives. At week 9, whether the layered system *beats* full history is judge-sensitive, and we report it as such: under the primary judge, hybrid v2 leads 93.2% vs. 88.9% (13/27 discordant, item-level $p = 0.039$; per-user effects $+3.7, -1.9, +7.4, +5.6, +7.4, +3.7$ pp, five of six positive, sign-flip $p = 0.063$); under the complete cross-family re-judging, full history edges ahead at week 9 (272 vs. 260 of 324, $p = 0.18$) and leads the tenure aggregate under that judge (79.7% vs. 72.4%; Appendix B), while under the primary judge the tenure aggregate is effectively tied (85.7% vs. 86.4%). The judge-robust statement is that *no judge-independent accuracy winner emerged at nine weeks, while the layered system reads roughly half the tokens per question* — with read cost growing far more slowly than full history's (below); the graph likewise shows no significant difference from full history under either judge. (Non-significance is not equivalence: we claim the absence of a robust winner, not demonstrated parity.) The recency window fails both regimes (42.9–66.4%; early-chapter 17/54 at week 9). Remaining baselines for future work: raw-chunk RAG with a competitive reranker, and an oracle-retrieval ceiling.

Cost. Marginal read cost is small: 677 (vector) to 1,807 (hybrid v2) input tokens per question — input-token charges of ≈$0.002/question at July-2026 list pricing for the answerer (input tokens only; output tokens, storage, retrieval infrastructure, and amortized write cost are excluded, and per-call wall-clock times are in the released logs rather than analyzed here); full history spends the entire corpus per question (∼2.8–5.2K tokens here), growing linearly with corpus size, with no judge-independent accuracy advantage to show for it (above); hybrid v2's read cost also grows (+48% by week 9) but from a base roughly half full history's at week 9. Two structural results: the budgeted map's read cost is flat with tenure (1,183 → 1,182 tok/q) while the unbounded winners' grows (+48–57% by week 9) — forgetting is what keeps the map cheap, and the tenure crossover is the price of that economy; and write-side quality is bought with tokens (the curated writer spends ∼4.6× the extractor's tokens to halve the audited error count) — but write cost amortizes over every future read, so the economics and the write-quality association (§5.5) point at the same place: the write path.

6 From findings to artifact: Veracium

The winning layered design is released as Veracium (`pip install veracium`; MIT), a provenance-aware memory library whose design decisions trace one-to-one to measured findings. The store is a typed graph with provenance-carrying edges (§5.6's supersede-never-erase, with prior values rendered as inline prose rather than versioned-edge dumps); raw episodes are deterministic per-event records; an LLM-curated, budget-aware wiki is compiled *above* the store (§5.3, §5.5); and an assertability gate implements the missing mechanism §5.7 specifies: facts whose only support is third-party-authored evidence are never assertable — they surface, if at all, as explicitly fenced unverified claims — and the gate extends through mixed-provenance events (content *derived from* third-party sources is trust-capped at the weaker source, closing the summarization-laundering channel that Louck (2026) formalizes — a channel independently discovered by Veracium's first production consumer on a real mailbox, and fixed at the representation level).

A distilled acceptance eval derived from the study's harness ships as the library's regression gate (small fictional scenarios; asserts zero injection assertions and a correctness floor), the full harness is released alongside the corpus generator (§Reproducibility), and an opt-in robustness tier replays real chat corpora through the write path, programmatically asserting quarantine, isolation, and provenance invariants. An internal robustness smoke test replayed 200 conversations (371 user turns) sampled with a fixed seed from a public chat corpus (LMSYS-Chat-1M; license-gated, not

redistributed — the shipped tier uses synthetic fixtures), with synthetic injection content inserted into 15% of turns and per-user isolation probed across all users; the programmatic checkers reported zero cross-user leaks, zero injection leaks, and zero malformed edges over all turns and probe answers (and, in earlier iterations, caught two real crash bugs — the eval-driven loop working as intended). This is an internal smoke test with scripted, non-blind detection — not a product evaluation or evidence of production-grade robustness; Veracium is the paper's artifact, not its subject.

7 Threats to validity

Synthetic corpus. Deliberate: real-entity benchmarks leak to pretraining (Xu et al., 2026, 2024), and truth-first generation is what makes gold answers script-valid by construction with validity intervals no real corpus offers. The residual is external validity: generated users are tidier than real ones, the renderer's messiness knobs were not pushed, and results should be read as architecture comparison under controlled conditions; a real-user study is future work.

LLM judge. The judge is versioned, treated as part of the benchmark, and held fixed across backends, so *shared* error cancels in differential claims (Zheng et al., 2023; Wang et al., 2024). That does not cover judge error that interacts with answer style — verbosity, abstention wording, prose vs. graph-derived evidence — which is backend-correlated, and matters most where margins are thin (the hybrid-v2–map short-horizon interval starts at +0.2 pp). Judge error is nonzero and measured only anecdotally (3 errors caught in ∼200 hand-spot-checked verdicts). As mitigation we ran a **complete cross-family re-judging**: every answer for every headline backend, the full-history baseline, and (in a follow-up pass) the vector backend — 8,985 answers across both corpora, with no selection on the original verdicts — was independently re-judged by DeepSeek v3.2, a different model family from answerer and judge. Overall agreement was 87.9% (Cohen's $\kappa = 0.60$); the disagreements are strongly asymmetric — 1,029 Sonnet-correct/DeepSeek-wrong vs. 54 the reverse — i.e., DeepSeek is systematically stricter (absolute scores ∼8–12 pp lower; per-judge aggregates and confusion counts in Appendix B), quantifying why we treat absolute levels as judge-defined throughout. Under DeepSeek-only verdicts: **judge-robust** — the graph–map crossover (week-3 endpoint $p = 0.010$, week-9 endpoint $p < 10^{-5}$, interaction positive for all six users, exact sign-flip $p = 0.031$), hybrid v2 > graph short-horizon ($p = 0.018$), and full history tying-or-beating the memory systems at the short horizon; **judge-sensitive** — the hybrid-v2–map short-horizon margin (direction preserved, $p = 0.15$) and the week-9 hybrid-v2–full-history comparison (direction reverses; §5.8) — both reported as such where they appear. A blinded *human* adjudication (pack prepared) remains the arbiter for the judge-sensitive pair before archival submission. The corpus generator (Nova Pro) is a different model family from the answerer (Haiku) and judge (Sonnet); the answerer and judge share a family.

Benchmark-informed iteration. Hybrid v2 was designed after observing hybrid v1's failure and the backend results on the same short-horizon corpus, so its 96.8% is partly a development-set number and is labeled as such (§5.3). Two facts limit the damage: the six long-horizon users were generated *after* hybrid v2 was fixed (new personas and preference chains; the architecture was not re-tuned for them), so the tenure result has a meaningful held-out character; and the released, seeded generator makes a fully hidden test corpus cheap to produce — which, together with a run on an external benchmark, is the planned pre-archival hardening.

Uncertainty and clustering. Headline comparisons are reported with exact paired McNemar tests and user-cluster bootstrap intervals (§5.3, §5.4); questions within a user share facts, so users are the clustering unit, and item-level McNemar p-values should be read as optimistic relative to cluster-level inference. The tenure analysis has only six user clusters: bootstrap coverage at that size is discrete, which is why §5.4 also reports all per-user interaction effects and exact sign-flip randomization — and why the graph–map interaction sits at that test's resolution floor ($p = 0.063$). Category-level comparisons (some decided by one or two answers) are descriptive only, and we flag the week-6 "tie" as a 2-answer gap rather than an estimate of equality. Per-user and per-replicate distributions are reported in Appendix A.

Single answerer. All comparisons fix one answering model — exactly the control MemDelta shows is necessary for interpretable comparisons (Wang, 2026), but it scopes generality: the write-path and topology findings are shown for one answerer. A second answerer on the tenure subset is the highest-value robustness addition and is planned.

Sample structure. The crossover is replicated at 6 users × 3 replicates (n=324/cell). Each user is an independent generator draw (persona and full preference chain); the three stochastic replicates replicate answer/judge stochasticity over the fixed corpora, not fresh content. Chapter-level work facts, however, are derived by chapter index, so same-archetype users share that skeleton, and the archetype draw was concentrated (4/2/0 across three archetypes). The inversion is robust across the variation present; per-replicate work-fact draws and archetype balancing are the cleanest next instrument fix.

Injection realism. All injection material is planted and template-grade, in a benign harness: 14 unique probe questions (one per user, evaluated over three stochastic trials each — 42 answer trials) plus the six-rung ladder — the documented attack class (Greshake et al., 2023; Chen et al., 2024), not adaptive adversaries, and not the laundering constructions of Louck (2026) (one of which our episodic-leak finding independently exhibits). Zero failures on 14 unique probes bounds the per-probe assertion rate only below ∼19% (95%, independence assumed — optimistic, as probes share a template family). Generating substantially more unique probes across multiple attack templates would strengthen this result more than further stochastic reruns of the same prompts. "Resists the attack class" is the claim; "secure" is not. Security must also be read jointly with clean third-party recall: the answerability stage excluded five questions precisely because blanket third-party distrust suppresses legitimate received information (§3.3) — a system can look safe partly by refusing; the graded-trust experiment (§5.7) is the start of quantifying that trade-off.

Short-horizon compression. At ∼30–60 records/user the corpus sits below ConvoMem's full-context threshold (Pakhomov et al., 2025), compressing short-horizon deltas between backends — no longer only an inference: our full-history baseline ties the best memory system in that regime (§5.8). This is precisely why the tenure results carry more weight than the bake-off, and why we report both. Raw-chunk RAG and an oracle-retrieval ceiling remain unrun.

8 Conclusion and future work

We inverted the benchmark pipeline — truth before text — and ran memory long. The instrument's features (validity intervals, trust-typed channels, benign injection probes, as-of-date sets) surfaced phenomena the standard pipeline cannot see: a replicated ranking inversion with history length,

an association between write-stage errors and downstream misses aligned fact-for-fact, injection resistance that tracked whether provenance boundaries survive representation, and a mismatch between overall accuracy and conditional abstention behavior. The layered architecture these findings select is released as an open-source library with the harness as its regression suite.

Future work, in priority order: an evidence-grounded abstention-gate evaluation (the mechanism §5.7 motivates; the artifact implements a first version); a second answerer on the tenure subset; a run on LongMemEval's update/abstention tracks to place the design in the field's comparison set; adaptive and laundering-grade injection (bridging to Louck, 2026; Dash et al., 2026); per-replicate work-fact draws and archetype balancing in the generator; compaction-loss and expiry-driven-confirmation experiments; and a real-user longitudinal study.

Reproducibility statement

We release: the corpus generator (deterministic, seeded; MIT), the full harness with per-cell model configurations, corpus-generator seeds, and stochastic-replicate identifiers (MIT), the synthetic corpus (CC BY 4.0), all judged verdicts (both judges), the complete per-call logs (model identifiers, tags, token counts; decoding at provider defaults), and Veracium (MIT). Exact model identifiers appear in §3.4. The Veracium library is public at `https://github.com/veracium-ai/Veracium`; the study-artifact package (generator, harness, corpora, verdicts, logs) is in preparation at the time of this preprint and will accompany the archival version with an immutable release tag and an archived snapshot DOI. The paper's numbers re-derive from the artifacts by script. The real-chat robustness corpus (LMSYS) is license-gated and is *not* redistributed; the robustness tier ships with synthetic fixtures instead, and nothing in the released corpus derives from LMSYS data.

Ethics and data statement

The corpus is fully synthetic and fictionalized: no real personal data, no scraped conversations, no real entities; injection content is template-grade and defensive in intent (it tests quarantine, not attack generation). Third-party-authored text is treated as untrusted *by design*, which we believe is the safety-relevant default for deployed memory systems.

AI-collaboration disclosure

This work was conducted as a human-directed, AI-executed collaboration: the author set research questions, made all design decisions, and verified results; AI assistants (Claude-family models) executed experiments, drafted analyses, and wrote code under direction, with every paper-bound figure re-derived from raw artifacts by script before submission.

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A Per-user and per-replicate distributions

Pooled percentages compress the variation reviewers need to assess clustering; Tables 4 and 5 report the underlying distributions. Per-replicate short-horizon totals are stable (none 15.3/15.3/14.9%; map 93.8/94.2/94.5%; vector 89.5/91.6/92.0%; graph 92.0/92.4/95.3%; hybrid v1 87.6/88.7/92.0%; hybrid v2 96.7/98.5/95.3%).

Table 4: Short-horizon accuracy (%) per user, pooled over 3 replicates (n=57–60 judged answers per cell; dov/mara/renata have hand-built question sets, g1–g11 the generated 20-question set).

User	none	map	vector	graph	hyb. v1	hyb. v2
dov	15	83	88	82	80	**98**
g1	15	**95**	88	87	82	92
g2	16	93	91	93	91	**95**
g3	16	91	91	98	96	**100**
g4	15	92	**95**	92	88	**95**
g5	16	93	96	93	93	**100**
g6	16	98	96	96	89	**100**
g7	15	93	88	**98**	**98**	97
g8	16	95	93	**100**	93	98
g9	13	**98**	87	92	90	95
g10	15	**100**	88	93	92	97
g11	15	**98**	95	93	93	**98**
mara	15	**100**	90	98	93	98
renata	15	88	87	90	73	**93**

Short-horizon per-user baselines (full history / window, %): dov 98/80, g1 98/73, g2 96/70, g3 98/65, g4 100/68, g5 91/68, g6 100/65, g7 98/75, g8 100/74, g9 97/75, g10 98/62, g11 97/73, mara 98/90, renata 100/82.

B Cross-family judge: agreement and aggregates

Confusion over all 8,985 re-judged answers — five backends × 1,797 (Sonnet × DeepSeek): both-correct 6,827; Sonnet-correct/DeepSeek-wrong 1,029; DeepSeek-correct/Sonnet-wrong 54; both-wrong 1,075. Observed agreement 87.9%; Cohen's $\kappa = 0.600$. Agreement conditioned on the primary

Table 5: Tenure accuracy (%) per user and checkpoint, pooled over 3 replicates (n=54/cell). Cell format: map / vector / graph / hybrid v2.

User	week 3	week 6	week 9
L101	81/70/72/78	76/57/83/85	72/69/100/98
L102	87/80/78/87	83/74/80/87	76/76/83/83
L103	83/78/85/83	85/80/80/87	74/76/85/87
L104	76/70/78/83	69/59/81/89	80/61/81/93
L105	85/81/78/76	85/80/78/85	85/81/98/100
L106	74/72/65/74	80/63/72/81	83/85/94/98

Table 6: History-baseline tenure accuracy (%) per user (§5.8), pooled over 3 replicates. Cell format: full history / recency window. Per-user week-9 hybrid v2 − full history effects: $+3.7, -1.9, +7.4, +5.6, +7.4, +3.7$ pp (L101–L106); five of six positive, exact six-user sign-flip $p = 0.063$.

User	week 3	week 6	week 9
L101	81/56	87/46	94/69
L102	89/48	93/43	85/65
L103	85/52	87/37	80/65
L104	80/39	81/43	87/65
L105	83/50	89/41	93/67
L106	76/52	78/48	94/69

verdict: 86.9% among Sonnet-correct answers, 95.2% among Sonnet-wrong — the second judge rarely rescues an answer the primary judge rejected, and its disagreements are concentrated in added strictness. (The four-backend subset of the first pass alone: agreement 88.1%, $\kappa = 0.574$.)

Table 7: Aggregate accuracy (%) under each judge (same answers, both corpora), with per-backend confusion counts (both corpora pooled, n=1,797/backend): BC = both-correct, SO = Sonnet-only correct, DO = DeepSeek-only correct, BW = both-wrong.

	short (825)		tenure (972)		confusion			
Backend	Son.	DS	Son.	DS	BC	SO	DO	BW
curated map	94.2	82.2	79.7	67.2	1322	230	9	236
vector	91.0	78.8	72.9	61.8	1241	219	10	327
graph	93.2	81.2	81.8	70.6	1353	211	3	230
hybrid v2	96.8	84.5	86.4	72.4	1392	247	9	149
full history	97.9	93.0	85.7	79.7	1519	122	23	133

Abstention classifier. The abstention/wrong-specific split uses a deterministic regular expression over the answer text, applied only to answers the judge marked wrong: `don't have | no (information|record|mention|notes) | don't know | not in (my|the) notes | no such | unable to` (case-insensitive). An answer matching anywhere is classified as an abstention, otherwise as a wrong specific — so a hedged answer that both abstains and asserts counts as an abstention, and partial answers are classified by the presence of any disclaimer phrase, a known coarseness. The classifier is reproducible but has not been validated against human labels; the planned blinded adjudication uses a three-way rubric (correct / wrong-specific / abstention) partly to measure this

classifier's agreement with human judgment.

Confabulation metrics under both judges (tenure; wrong-set defined by each judge): conditional confident-error share — vector 62.7% (Sonnet) / 62.5% (DeepSeek); map 72.1 / 69.3; graph 83.6 / 78.0; hybrid v2 93.9 / 85.4. Unconditional wrong-specific rate — vector 17.0 / 23.9; map 14.6 / 22.7; graph 15.2 / 22.9; hybrid v2 12.8 / 23.6; full history (DeepSeek) 18.1. The conditional ordering (hybrid v2 most confident when wrong, vector most abstaining) is stable across judges; the unconditional ordering is judge-sensitive.